

# ISP Training Program – AMR DevKit and Perceptor

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# Agenda

- Brief Introduction to AMR
- Overview of AMR DevKit
- Robot Perception
  - AMR DevKit: Bring Up Sensors on Robots
  - AMR Perceptor: Sensing to Reasoning
- Demo
  - AMR Assembly, Bring Up Sensors, Control AMR
  - Run AMR Perceptor

# ISP Training Program

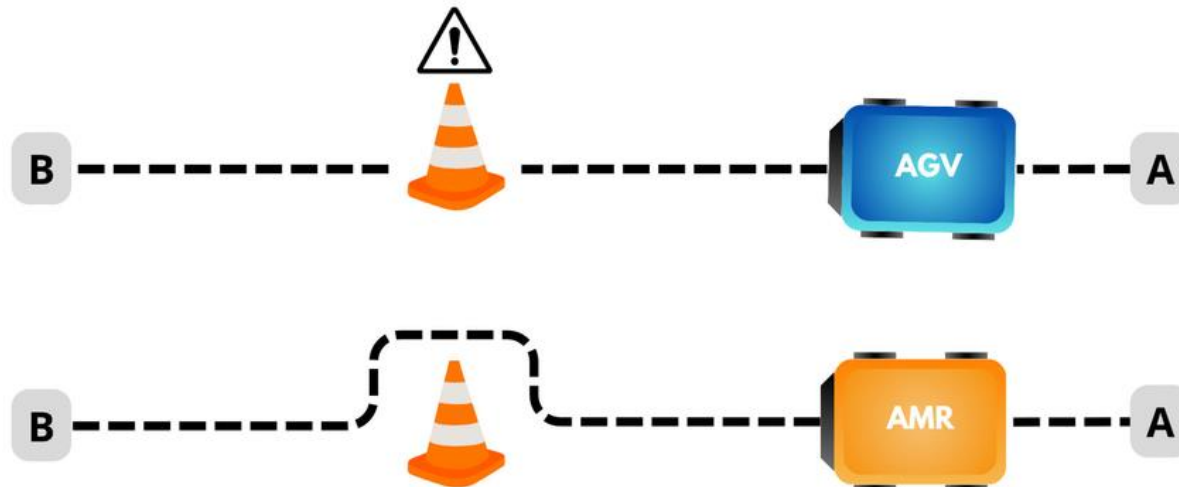
| Training Date                      | Projects                    | Hardware                | NVIDIA Base  | GitHub                                   |
|------------------------------------|-----------------------------|-------------------------|--------------|------------------------------------------|
| 2026.04.15 (en)<br>2026.04.22 (zh) | Edge Agent                  | MIC-733                 | Agent Studio | <a href="#">LINK</a>                     |
| 2026.04.29 (en)<br>2026.05.06 (zh) | VSS on Jetson Thor          | MIC-742                 | VSS 2.4.1    | <a href="#">LINK</a>                     |
| 2026.08.19 (zh)                    | AMR DevKit<br>AMR Perceptor | AMR DevKit<br>(MIC-732) | -            | <a href="#">AMR DevKit<br/>Perceptor</a> |
| TBD                                | AMR SLAM Navigator          | AMR DevKit<br>(MIC-732) | -            | <a href="#">LINK</a>                     |
| TBD                                | AMR ReMEmbR                 | AMR DevKit<br>(MIC-732) | ReMEmbR      | <a href="#">ReMEmbR</a>                  |
| TBD                                | Omniverse/Issac Sim         | -                       | -            | -                                        |

# Brief Introduction to AMR



# Mobile Robots

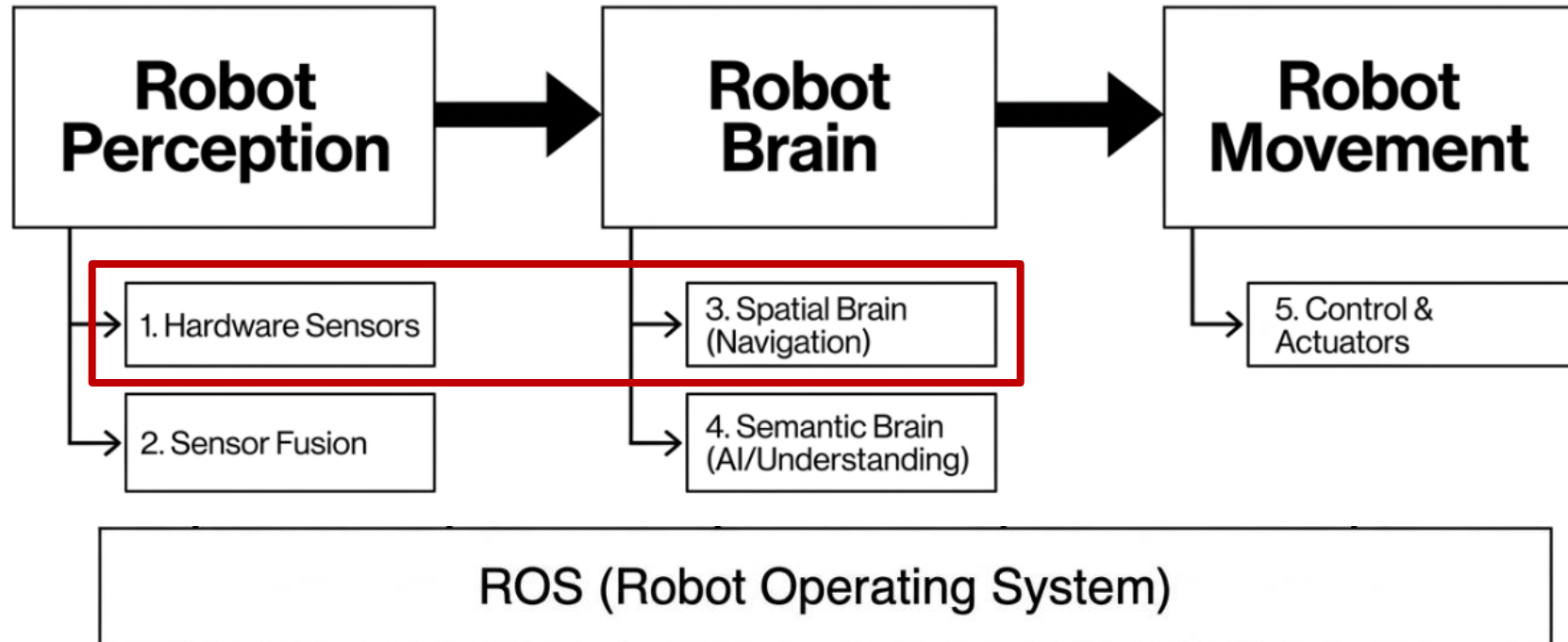
- Common types of mobile robots:
  - Autonomous Guided Vehicles (AGVs)
  - Autonomous Mobile Robots (AMRs)
  - Mobile Manipulator Robots (MMRs)
  - Humanoid Robot (Legged)



# AGVs vs. AMRs

- AGVs
  - Follow fixed path and predetermined routes
  - Often require modifying environment (tape, wires, or rails)
  - Follow same path to transport an object
- AMRs
  - Navigation system with mapping, localization abilities
  - Sensors for detecting and avoiding obstacles
  - Path planning

# High-level System Design of Robot



For Advantech AMR DevKit,

- It is integrated with all required sensors and be ready for SLAM
- SLAM (Simultaneously Localization and Mapping) will be introduced in upcoming session



# Overview of AMR DevKit





# Advantech AMR DevKit

- Hardware
  - MIC-732 for computing
  - Sensors (stereo cameras, 2D LiDAR, fisheye cameras)
  - Power supply (battery included and power supply unit for AC)
- Software
  - JetPack 6.0 with ROS package installed
  - Sensor drivers included
  - Support other ISP packages (available on GitHub)
    - AMR Perceptor and AMR ReMEmbR

# AMR DevKit: Front Side



# AMR DevKit: Rear Side



# Light Indicator

- White (blinking): starting
- Light blue (blinking): standby
- Blue: operating
- Yellow: stop operating
- Light orange: low battery
- Red: error

# Robot Perception

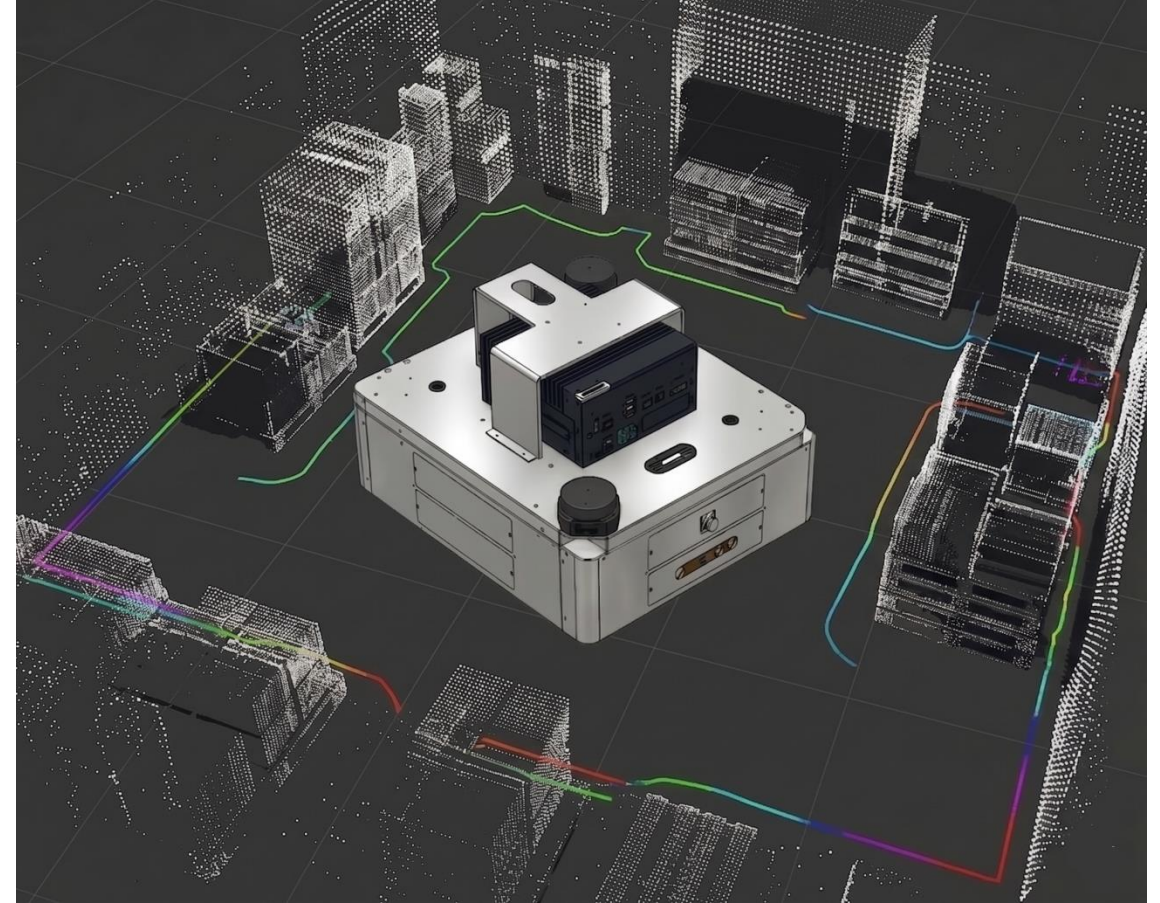


# Sensors on AMR DevKit

- Cameras
  - Stereo Camera  $\times 2$ 
    - Measure depth or distance of objects
    - The output is a point cloud
  - Fisheye Camera  $\times 2$ : not used by AMR DevKit
- 2D LiDAR (Light Detection and Ranging)  $\times 2$ 
  - Two of them are placed diagonally
  - Operate on a single horizontal
  - Emit beams in a fan-like pattern to detect surrounding

# Bring Up Sensors

- 3D point cloud of surrounding obstacles, dynamically generated in real-time by stereo cameras
- 2D LiDAR scan profiles mapping the environment boundaries for robot navigation



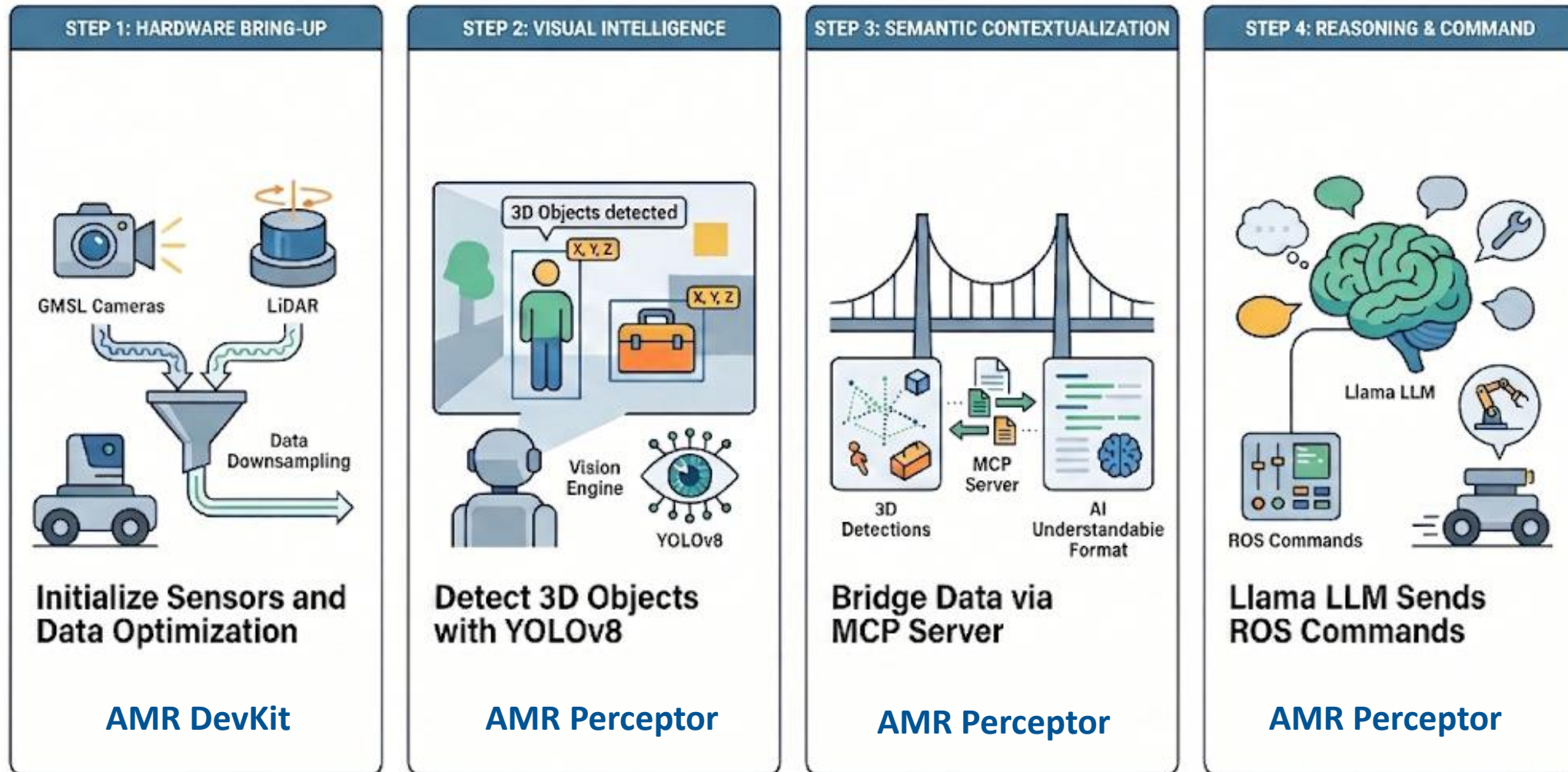
This is a conceptual illustration (AI-generated), not a real screenshot.



# AMR Perceptor

- No wheelbase is needed
- Project for demonstrating:
  - Object detection with YOLO: recognizing surrounding objects
  - LLM-driven AMR control via WebUI chat interface

# Reasoning Loop



**ADVANTECH**

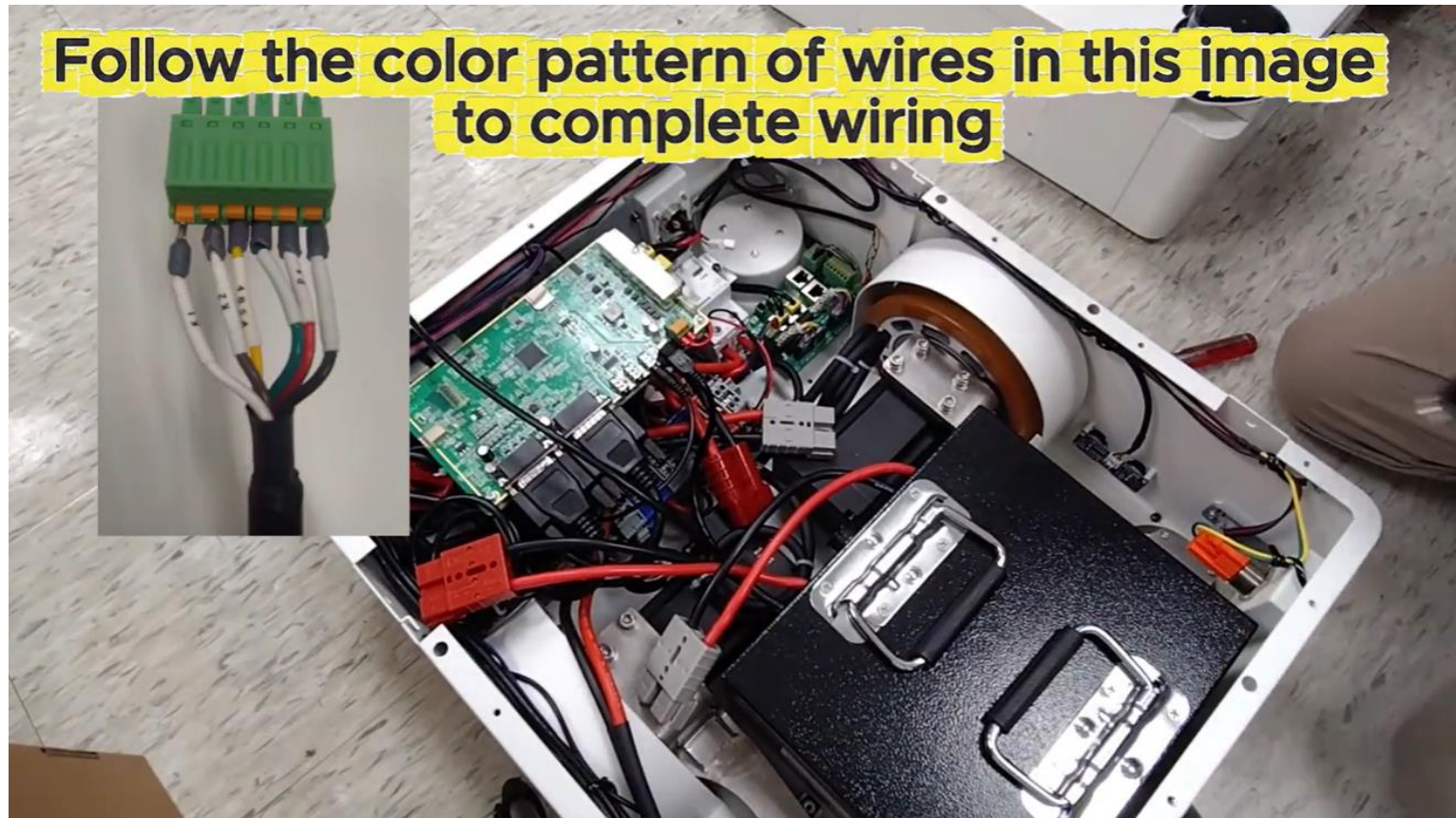
*Enabling an Intelligent Planet*

# Demo



# AMR Assembly

- Watch [video tutorial](#)





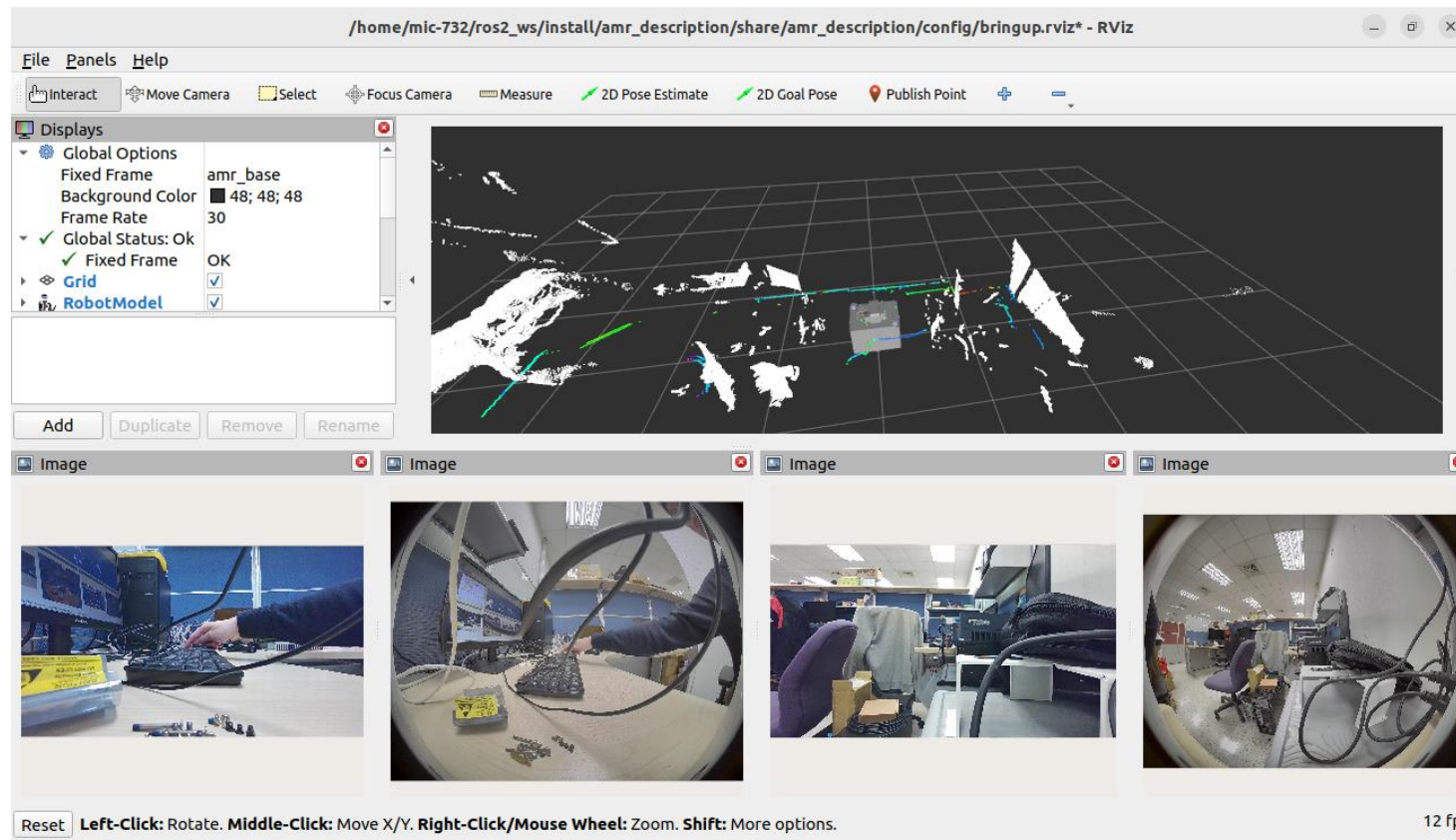
# Accessories

- Portable monitor with VESA wall-mounting compatibility
  - Provides live preview of cameras and LiDARs
- PS4 controller
  - Provides direct control for visitors in demo



# Bring Up Sensors

- Follow [Installation Guide](#) to install and setup your AMR
- Bring up sensors:
  - `ros2 launch amr_description bringup.launch.py bringup_rviz:=true`



# Control AMR with Keyboard

- Connect keyboard with MIC-732 via USB cable
- Enter commands:
  - `cd ~/ros2_amr100`
  - `source docker/scripts/deploy_arm64.sh`
  - `docker exec -u adata -it ros_dev-aarch64-container bash`
  - `source env_setting.sh 99`
  - `ros2 run teleop_twist_keyboard teleop_twist_keyboard`
- Watch [video tutorial](#)
- Key mapping

|                   |                    |                    |
|-------------------|--------------------|--------------------|
| u (Forward Left)  | i (Forward)        | o (Forward Right)  |
| j (Rotate Left)   | k (Emergency Stop) | l (Rotate Right)   |
| m (Backward Left) | , (Backward)       | . (Backward Right) |



# Control AMR with PS4 Controller

- (Terminal 1)
  - `cd ros2_amr100source docker/scripts/deploy_arm64.sh`
  - `docker exec -u adata -it ros_dev-aarch64-container bash`
  - `source env_setting.sh 99`
  - `ros2 run joy joy_node`
  - Expected outcome:
    - `[INFO] [1786673510.830800078] [joy_node]: No haptic (rumble) available, skipping initialization`
    - `[INFO] [1786673510.831036798] [joy_node]: Opened joystick: PS4 Controller. deadzone: 0.050000`
- (Terminal 2)
  - `docker exec -u adata -it ros_dev-aarch64-container bash`
  - `source env_setting.sh 99ros2 run teleop_twist_joy teleop_node --ros-args \`
  - `-p axis_linear.x:=1 \`
  - `-p scale_linear.x:=0.2 \`
  - `-p axis_angular.yaw:=0 \`
  - `-p scale_angular.yaw:=0.5 \`
  - `-p enable_button:=9`
  - Expected outcome:
    - `[INFO] [1786674169.875979616] [TeleopTwistJoy]: Teleop enable button 9.`
    - `[INFO] [1786674169.876064576] [TeleopTwistJoy]: Linear axis x on 1 at scale 0.200000.`
    - `[INFO] [1786674169.876084928] [TeleopTwistJoy]: Angular axis yaw on 0 at scale 0.500000.`

# Control AMR with Mobile App

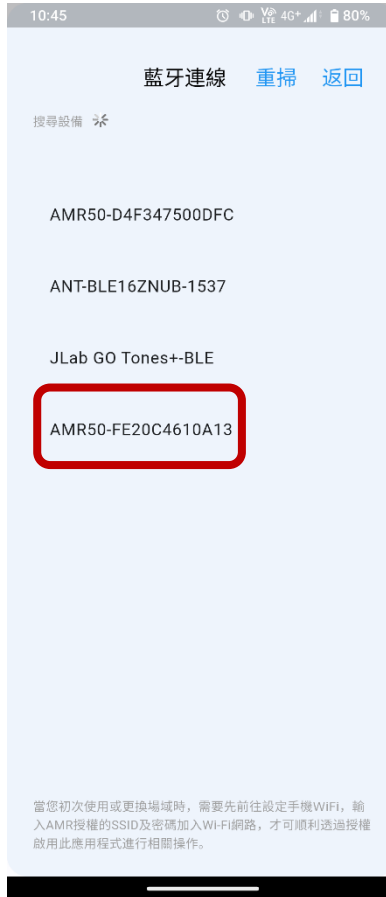
- Chinese interface only
- iOS:
  - Download the app from App Store
- Android:
  - Download the APK from Google Drive and install it manually

# Control AMR with Mobile App

## Add AMR



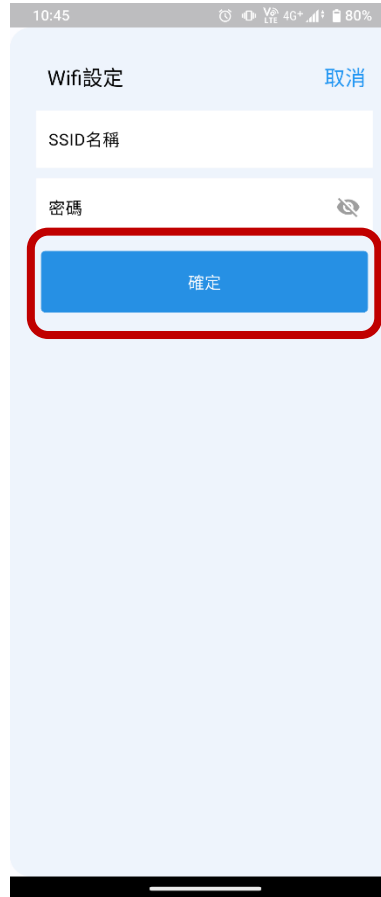
## Select your AMR



## Continue



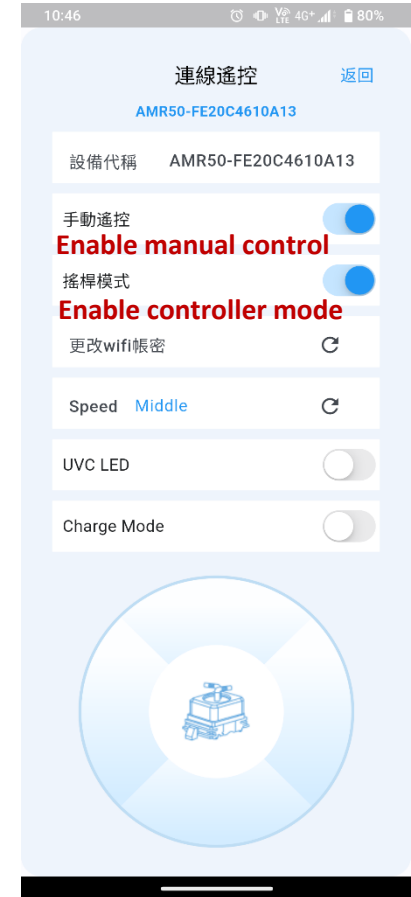
## Continue



## Connect your AMR



## Control your AMR



# Run AMR Perceptor

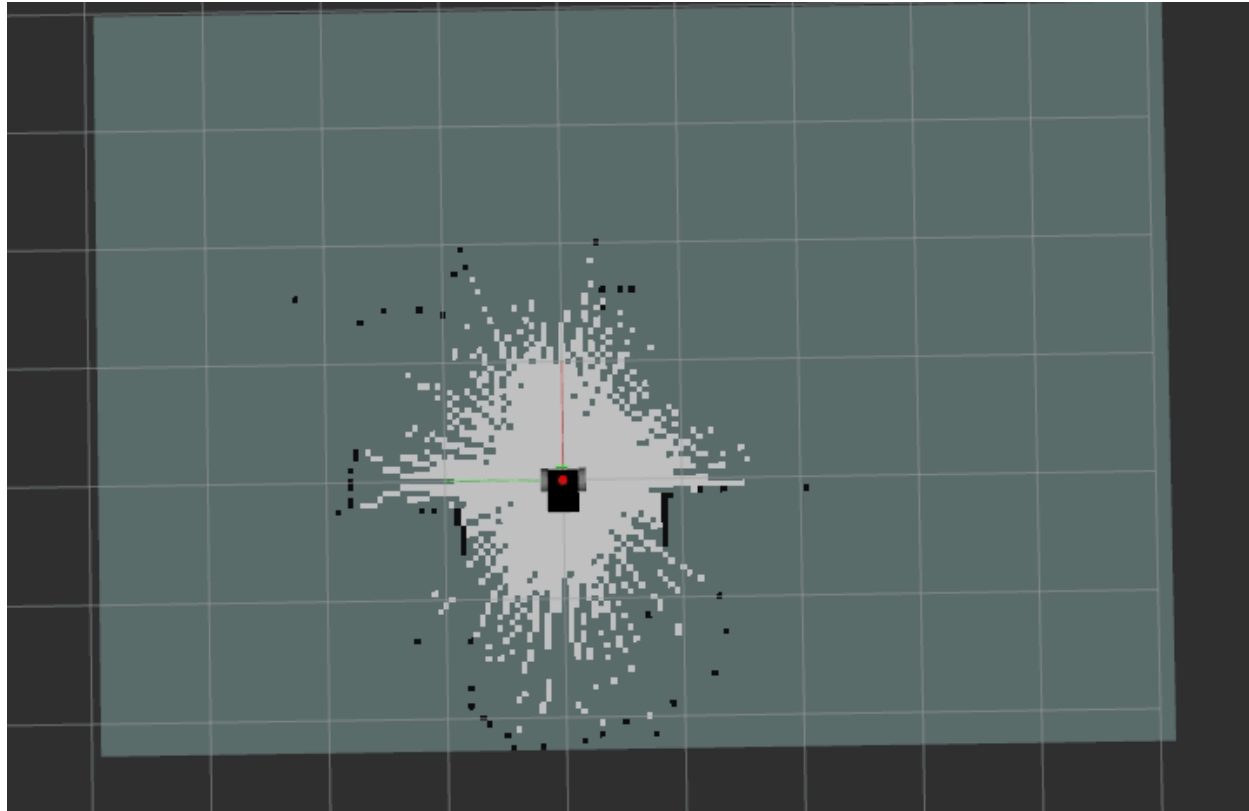
- Start YOLO, MCP, llama.cpp, and Web UI
  - `cd AMR_Perceptor`
  - `docker compose up -d`
- See perception with YOLO detection in Rviz
  - `ros2 launch amr_description bringup.launch.py yolo_rviz:=true preprocess:=true`
- Watch [video tutorial](#)
- For the usage of Web UI, please refer to instructions on GitHub

# Follow Instructions on GitHub

- AMR DevKit
  - [https://github.com/advantech-EdgeAI/AMR\\_DevKit](https://github.com/advantech-EdgeAI/AMR_DevKit)
- AMR Perceptor
  - [https://github.com/advantech-EdgeAI/AMR\\_Perceptor](https://github.com/advantech-EdgeAI/AMR_Perceptor)
- Any questions? Go to the projects' GitHub Issues

# Next Step

- Next time, we will drive AMR to build a map in real world
  - New skill will be unlocked: SLAM





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